



Sinclair ZX81 / ZX80 — Pinouts

Z80A · ULA · 2114 RAM · ROM

DIP pinouts from the top, notch up, pin 1 top-left. Power **red**, ground grey, clocks blue. Verify against the datasheet before powering up.

Legend — reading the pinouts notation

/NAME Active LOW — true / asserted when the pin is at logic **LOW** (~0 V). A leading / (datasheets often use an overline) marks these: e.g. /RES /IRQ /CS /WE.

NAME Active HIGH / normal — asserted when the pin is **HIGH** (~+5 V). No slash. R/W is a level: HIGH = read, LOW = write.

Colour key: **Vcc / Vdd = power** · **GND / Vss = ground** · **φ = clock**. **Orientation:** chip seen from the top, notch up — pin 1 top-left, numbers run **down the left** then **up the right**.

Z80A CPU

40-DIP

A11	1	40	A10
A12	2	39	A9
A13	3	38	A8
A14	4	37	A7
A15	5	36	A6
CLK	6	35	A5
D4	7	34	A4
D3	8	33	A3
D5	9	32	A2
D6	10	31	A1
Vcc	11	30	A0
D2	12	29	GND
D7	13	28	/RFSH
D0	14	27	/M1
D1	15	26	/RESET
/INT	16	25	/BUSRQ
/NMI	17	24	/WAIT
/HALT	18	23	/BUSAK
/MREQ	19	22	/WR
/IORQ	20	21	/RD

ZX81 CPU @ 3.25 MHz. The ULA off-loads the CPU during display (NMI generator). **Fails:** dead machine, no boot.

ULA 2C184/2C210 gate array

40-DIP

A7'	1	40	VCC
A8'	2	39	A6'
A2'	3	38	A5'
A1'	4	37	A4'
A0'	5	36	A3'
/RD	6	35	OSC
/IORQ	7	34	GND
/WR	8	33	KBD0
/MREQ	9	32	D0
/M1	10	31	KBD1
A14	11	30	D1
/RAMCS	12	29	KBD2
/ROMCS	13	28	D2
/CLK	14	27	KBD3
/NMI	15	26	D3
TV - TAPE	16	25	KBD4
/HALT	17	24	D4
A15	18	23	D5
D7	19	22	UK/US
TAPE IN	20	21	D6

Ferranti ULA: video sync, character access, keyboard, tape, NMI generator. Different chip from the Spectrum ULA. Pinout verified (D7 on pin 19; D0-D6 on the right). **Fails:** no display, no boot.

2114 SRAM 1K×4

18-DIP

A6	1	18	Vcc
A5	2	17	A7
A4	3	16	A8
A3	4	15	A9
A0	5	14	D1
A1	6	13	D2
A2	7	12	D3
/CS	8	11	D4
GND	9	10	/WE

1KB RAM = two 2114 (4-bit) on the base ZX81 (16K via the wobbly RAM pack). **Fails:** no/garbled listing, crashes.

2716/2316 ROM 2K×8

24-DIP

A7	1	24	Vcc
A6	2	23	A8
A5	3	22	A9
A4	4	21	Vpp
A3	5	20	/OE
A2	6	19	A10
A1	7	18	/CE
A0	8	17	07
00	9	16	06
01	10	15	05
02	11	14	04
Vss	12	13	03

8K BASIC ROM (mask ROM; shown as the generic 2716/2364 family). **Verify exact ROM pinout.**

Test & Repair

tips

No display ULA / Z80 / crystal

No boot Z80 / ROM

RAM-pack crashes edge connector (clean & reseal)

Garbage screen 2114 RAM / ULA

The single +5V rail simplifies things; the ULA and the RAM-pack connector are the usual culprits.

Notes

△ verify these

BASIC ROM exact part (8K mask vs the 2K family shown)

(Z80, ULA, 2114 verified)

Amber pins and △ **flagged chips** are my best reconstruction from incomplete/garbled sources — confirm each against a datasheet before use.